Shahariar Ifti

Research Assistant & Data Scientist

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Research Interests

Human-Computer Interaction; wearable and physiological sensing (HRV, GSR, IMU) for stress, mental health and rehabilitation; machine learning on physiological signals that generalizes to unseen individuals; low-cost health technology and human-centered design for resource-constrained communities in the Global South.

Education

- **North South University** — Dhaka, Bangladesh
  Bachelor of Science - Computer Science & Engineering; CGPA: 3.44/4.00 — *Dec 2024*
  Courses: *Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems Design, Database Systems, Machine Learning, Pattern Recognition and Neural Network, Theory of Computation*
- **Notre Dame College** — Dhaka, Bangladesh
  Higher Secondary Certificate (HSC) - Science; GPA: 5.00/5.00 — *2018*
- **Monipur High School** — Dhaka, Bangladesh
  Secondary School Certificate (SSC) - Science; GPA: 5.00/5.00 — *2016*

Publications

Published

- **Spondon: A Low-Cost, Multi-Sensor Wearable for Understanding Stress in Resource-Constrained Countries** — ACM COMPASS '26
  Shahariar Ifti*, R. J. Rony, S. Chakraborty, A. Mahbub, N. Ahmed* — *July 2026 | View Paper*
  Summary: *Designed an ESP32-based multi-sensor wearable (MAX30102, GSR, temperature) from locally available sensors to monitor stress in resource-constrained environments. Evaluated on 15 participants using the Stroop test; statistical analysis confirmed a 7.95% drop in HRV during stress. Using leave-one-subject-out (LOSO) cross-validation to test generalization to unseen individuals, a Random Forest classifier with personalized baseline normalization achieved 75.0% accuracy in binary acute stress detection and 58.4% in multi-class continuous monitoring.*

Under Review

- **"My Mind Is a Combination of My Body, My Money, and My Study": Contextualizing Mental Health Support Ecologies for Tertiary Students in Bangladesh** — ACM CHI '27
  Co-author — *Under Review*
- **SHONCHALON: A Wearable for Real-Time Multi-Class Exercise Recognition and Feedback in Post-Stroke Upper-Limb Rehabilitation** — IEEE PerCom '27
  Co-author — *Under Review*

In Preparation

- **Technology adoption among low-literate women (Co-learning & Familiarization Project)** — Target: ACM IMWUT
  Manuscript in preparation
- **Long-term stress among physicians during the COVID-19 pandemic** — Target: JMIR Mental Health
  Manuscript in preparation, with Cardiff University, UK

Experience

- **Research Assistant & Data Scientist (Full-time)** — Design Inclusion and Access Lab (DIAL), NSU
  Supervisor: Dr. Nova Ahmed — *June 2024 – Present*
  HCI research on health and wellbeing: developing custom wearable hardware, collecting and analyzing physiological health data (HRV, GSR, IMU) with ML and DL models, conducting literature reviews and mixed-methods studies, and contributing to publications.
- **Teaching Assistant** — DMP, NSU
  Jan 2023 – June 2023
  Led discussions, graded assignments, and mentored students to improve outcomes.

Research Projects

- **HEAT Subproject (World Bank funded)** — DIAL, NSU
  Research Assistant — *Jan 2026 – Present*
  Status: Ongoing; co-authored paper under review at ACM CHI '27. *Nationwide mixed-methods study assessing the mental health of tertiary-level students in Bangladesh and identifying technological solutions. Shortlisting participants via PSS/PHQ-9 scales for physiological stress monitoring and co-designing AI-driven interventions.*
- **Co-learning & Familiarization Project** — DIAL, NSU
  Research Assistant — *June 2025 – Present*
  Status: Ongoing; manuscript in preparation for ACM IMWUT. *Analyzing technology adoption among low-literate women through regional workshops. Developed custom hardware to measure physiological stress variability and implemented a family-based mentoring framework.*
- **Shonchalon: Rehabilitation of Post-Stroke Upper Limb Complications** — DIAL, NSU
  Research Assistant — *June 2024 – Oct 2025*
  Status: Under review at IEEE PerCom '27. *Developed a low-cost wearable using an ESP32-BLE and ICM20948 IMU (accelerometer, gyroscope, and magnetometer) to recognize five essential rehabilitation exercises and assess the correctness of hand movements. Integrated a custom Android application via Bluetooth for real-time 5Hz data logging and analyzed the data.*

- **From Travel Experience to Mobility Evaluation: A Real-Time Behavioural Study of Urban Commuting in Dhaka** — DIAL, NSU & University of Leeds, UK
  Research Assistant, with Dr. Charisma F. Choudhury — *Aug 2026 – Present*
  Status: Ongoing. *Investigating the physiological stress of commuters by integrating smartphone-based experience sampling with continuous physiological data (Empatica EmbracePlus). Analyzing multi-modal data (heart rate, EDA, skin temperature, self-reported stress) to evaluate mobility and mode preferences.*
- **Long-term Stress among Physicians during COVID-19** — Cardiff University, UK
  Collaborator, with Dr. Rahat Jahangir Rony and Dr. Naeima Hamed — *Ongoing*
  Analyzing long-term stress among physicians during the COVID-19 pandemic; manuscript in preparation for JMIR Mental Health.
- **ChatGPT Use among Teenagers in Bangladesh** — Pennsylvania State University, USA
  Collaborator, project led by Dr. Tanusree Sharma — *Ongoing*
  Contributing to a prompt-based study of how teenagers in Bangladesh use ChatGPT.

Selected Projects

- **Ambient Environment and Sleep Time Monitoring Wearable** — Course Project (CSE323)
  github.com/Sh-Ifti/sleep-monitoring-wearable — *June 2024*
  Built a low-cost Arduino wrist wearable with an MPU6050 accelerometer, PPG pulse sensor and light sensor, streaming data over Bluetooth (HC-05). Estimated sleep duration from windowed movement analysis in Python and assessed the sleep environment from ambient light and heart rate.
- **Personalized Meal Recommendation System (OnnoMeal)** — Senior Design Project
  github.com/Sh-Ifti/onnomeal-meal-recommender — *Jan 2024 – Dec 2024*
  Developed a Flask web application that recommends meals from a database of 305 traditional Bangladeshi recipes, matching each user's calorie and macronutrient targets (Mifflin–St Jeor energy estimate, weighted nutrient-distance ranking). Tools: Python, Flask, pandas, JavaScript, MySQL.
- **Predicting Airbnb Listing Prices in Europe** — Machine Learning Project
  github.com/Sh-Ifti/airbnb-price-prediction-europe — *Feb 2024 – June 2024*
  Compared six regression models (Linear, Ridge, SVR, Decision Tree, Random Forest, Gradient Boosting) for predicting listing prices across ten European cities; Random Forest performed best (test $R^2 = 0.854$ vs. 0.656 baseline). Tools: Python, scikit-learn, pandas, NumPy, Matplotlib.
- **Flood Mapping of Bangladesh** — Junior Design Project
  github.com/Sh-Ifti/bangladesh-flood-mapping — *July 2023 – Dec 2023*
  Produced flood inundation maps of Bangladesh for 2020–2022 from Sentinel-1 SAR imagery in Google Earth Engine (UN-SPIDER change-detection method), plus statistical visualizations of historical flood data from the Bangladesh Water Development Board. Tools: JavaScript, Google Earth Engine.
- **Regulated Smart Device for Children** — Directed Research Project (CSE498R)
  Supervisor: Muhammad Shafayat Oshman, Lecturer, North South University — *June 2024 – Dec 2024*
  Investigated alternative solutions to mitigate smartphone and digital device addiction among children.

Skills Summary

- **Sensing & Hardware**: ESP32, Arduino, BLE, ICM20948 / MPU6050 IMU, MAX30102 (PPG), GSR, Empatica EmbracePlus
- **Data Science & ML**: HRV and GSR analysis, Random Forest, LOSO cross-validation, deep learning, statistical analysis, scikit-learn, pandas, NumPy, Matplotlib
- **Programming**: Python, C++, Java, JavaScript, SQL, HTML & CSS
- **Research Methods**: Mixed methods, user studies, workshops and co-design, PSS / PHQ-9 instruments
- **Tools**: Git, Linux, LaTeX, Google Earth Engine, MySQL, MS Office
- **Soft Skills**: Communication, Team Collaboration, Problem-Solving

Achievements

- Secured 1st place in Database Project Presentation Competition (ESRD Lab BUET & NSU ACM) - 2023

References

Available upon request.